

$$\begin{aligned} & \rightarrow \frac{1}{16\pi^2} \\ & \left(-\frac{1}{54} \sum_{\mathbf{p}} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{3,0} [\mathbf{m}_d^{\mathbf{p}}] + \frac{5}{144} \sum_{\mathbf{p}} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{4,-1} [\mathbf{m}_d^{\mathbf{p}}] - \frac{2}{135} \sum_{\mathbf{p}} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{5,-2} [\mathbf{m}_d^{\mathbf{p}}] - \frac{1}{18} \sum_{\mathbf{p}} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{3,0} [\mathbf{m}_e^{\mathbf{p}}] + \right. \\ & \frac{5}{48} \sum_{\mathbf{p}} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{4,-1} [\mathbf{m}_e^{\mathbf{p}}] - \frac{2}{45} \sum_{\mathbf{p}} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{5,-2} [\mathbf{m}_e^{\mathbf{p}}] - \frac{1}{144} \sum_{\mathbf{p}} (4 \mathbf{g}_1^4 + 9 \mathbf{g}_2^4) \mathbf{L} \mathbf{F}_{3,0} [\mathbf{m}_l^{\mathbf{p}}] + \\ & \frac{1}{96} \sum_{\mathbf{p}} (5 \mathbf{g}_1^4 + 6 \mathbf{g}_2^4) \mathbf{L} \mathbf{F}_{4,-1} [\mathbf{m}_l^{\mathbf{p}}] - \frac{1}{45} \sum_{\mathbf{p}} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{5,-2} [\mathbf{m}_l^{\mathbf{p}}] + \\ & \left. \left(\frac{3}{4} \mathbf{g}_2^2 \overline{\mathbf{y}}_u^{\text{pr}} \mathbf{y}_u^{\text{pr}} - \frac{1}{432} \sum_{\mathbf{p}} (4 \mathbf{g}_1^4 + 81 \mathbf{g}_2^4) \right) \mathbf{L} \mathbf{F}_{3,0} [\mathbf{m}_q^{\mathbf{p}}] + \right. \\ & \left(-\frac{3}{4} \mathbf{g}_2^2 \overline{\mathbf{y}}_u^{\text{pr}} \mathbf{y}_u^{\text{pr}} + \frac{1}{288} \sum_{\mathbf{p}} (5 \mathbf{g}_1^4 + 54 \mathbf{g}_2^4) \right) \mathbf{L} \mathbf{F}_{4,-1} [\mathbf{m}_q^{\mathbf{p}}] - \frac{1}{135} \sum_{\mathbf{p}} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{5,-2} [\mathbf{m}_q^{\mathbf{p}}] - \\ & \frac{2}{27} \sum_{\mathbf{p}} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{3,0} [\mathbf{m}_u^{\mathbf{p}}] + \frac{5}{36} \sum_{\mathbf{p}} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{4,-1} [\mathbf{m}_u^{\mathbf{p}}] - \frac{8}{135} \sum_{\mathbf{p}} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{5,-2} [\mathbf{m}_u^{\mathbf{p}}] - \frac{1}{36} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{3,0} [\tilde{\mu}] - \\ & \frac{1}{24} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{4,-1} [\tilde{\mu}] + \frac{2}{45} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{5,-2} [\tilde{\mu}] - \frac{1}{32} \mathbf{g}_1^4 \mathbf{L} \mathbf{F}_{2,2,-1} [\mathbf{m}_1, \tilde{\mu}] - \frac{1}{16} \mathbf{g}_1^4 \mathbf{m}_1^2 \mathbf{L} \mathbf{F}_{2,2,0} [\mathbf{m}_1, \tilde{\mu}] - \\ & \frac{17}{32} \mathbf{g}_2^4 \mathbf{L} \mathbf{F}_{2,2,-1} [\mathbf{m}_2, \tilde{\mu}] - \frac{1}{16} \mathbf{g}_2^4 \mathbf{m}_2^2 \mathbf{L} \mathbf{F}_{2,2,0} [\mathbf{m}_2, \tilde{\mu}] + \mathbf{g}_2^4 \mathbf{L} \mathbf{F}_{3,2,-2} [\mathbf{m}_2, \tilde{\mu}] - \\ & \frac{3}{4} \mathbf{g}_2^4 \mathbf{m}_2^2 \mathbf{L} \mathbf{F}_{3,2,-1} [\mathbf{m}_2, \tilde{\mu}] - \frac{1}{2} \mathbf{g}_2^4 \mathbf{L} \mathbf{F}_{3,3,-3} [\mathbf{m}_2, \tilde{\mu}] + \frac{1}{2} \mathbf{g}_2^4 \mathbf{m}_2^2 \mathbf{L} \mathbf{F}_{3,3,-2} [\mathbf{m}_2, \tilde{\mu}] - \\ & \frac{1}{2} \mathbf{g}_2^4 \mathbf{L} \mathbf{F}_{4,2,-3} [\mathbf{m}_2, \tilde{\mu}] + \frac{1}{2} \mathbf{g}_2^4 \mathbf{m}_2^2 \mathbf{L} \mathbf{F}_{4,2,-2} [\mathbf{m}_2, \tilde{\mu}] - \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{\mathbf{y}}_d^{\text{pr}} \mathbf{y}_d^{\text{pr}} \mathbf{L} \mathbf{F}_{2,2,0} [\mathbf{m}_d^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{p}}] + \\ & \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{\mathbf{y}}_d^{\text{pr}} \mathbf{y}_d^{\text{pr}} \mathbf{L} \mathbf{F}_{3,2,-1} [\mathbf{m}_d^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{p}}] - \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{\mathbf{y}}_e^{\text{pr}} \mathbf{y}_e^{\text{pr}} \mathbf{L} \mathbf{F}_{2,2,0} [\mathbf{m}_e^{\mathbf{r}}, \mathbf{m}_l^{\mathbf{p}}] + \\ & \frac{1}{12} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{\mathbf{y}}_e^{\text{pr}} \mathbf{y}_e^{\text{pr}} \mathbf{L} \mathbf{F}_{3,2,-1} [\mathbf{m}_e^{\mathbf{r}}, \mathbf{m}_l^{\mathbf{p}}] + \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{\mathbf{y}}_e^{\text{pr}} \mathbf{y}_e^{\text{pr}} \mathbf{L} \mathbf{F}_{3,1,0} [\mathbf{m}_l^{\mathbf{p}}, \mathbf{m}_e^{\mathbf{r}}] + \\ & \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{\mathbf{y}}_e^{\text{pr}} \mathbf{y}_e^{\text{pr}} \mathbf{L} \mathbf{F}_{3,2,-1} [\mathbf{m}_l^{\mathbf{p}}, \mathbf{m}_e^{\mathbf{r}}] - \frac{1}{8} \tilde{\mu}^2 \overline{\mathbf{y}}_e^{\text{pr}} \mathbf{y}_e^{\text{pr}} (3 \mathbf{g}_1^2 + \mathbf{g}_2^2) \mathbf{L} \mathbf{F}_{4,1,-1} [\mathbf{m}_l^{\mathbf{p}}, \mathbf{m}_e^{\mathbf{r}}] + \\ & \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{\mathbf{y}}_e^{\text{pr}} \mathbf{y}_e^{\text{pr}} \mathbf{L} \mathbf{F}_{5,1,-2} [\mathbf{m}_l^{\mathbf{p}}, \mathbf{m}_e^{\mathbf{r}}] + \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{\mathbf{y}}_d^{\text{pr}} \mathbf{y}_d^{\text{pr}} \mathbf{L} \mathbf{F}_{3,1,0} [\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_d^{\mathbf{r}}] + \\ & \frac{1}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{\mathbf{y}}_d^{\text{pr}} \mathbf{y}_d^{\text{pr}} \mathbf{L} \mathbf{F}_{3,2,-1} [\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_d^{\mathbf{r}}] - \frac{1}{8} \tilde{\mu}^2 \overline{\mathbf{y}}_d^{\text{pr}} \mathbf{y}_d^{\text{pr}} (5 \mathbf{g}_1^2 + 3 \mathbf{g}_2^2) \mathbf{L} \mathbf{F}_{4,1,-1} [\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_d^{\mathbf{r}}] + \\ & \frac{1}{2} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{\mathbf{y}}_d^{\text{pr}} \mathbf{y}_d^{\text{pr}} \mathbf{L} \mathbf{F}_{5,1,-2} [\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_d^{\mathbf{r}}] - \frac{3}{4} \overline{\mathbf{y}}_u^{\text{pr}} \overline{\mathbf{y}}_u^{\text{st}} \mathbf{y}_u^{\text{pt}} \mathbf{y}_u^{\text{sr}} \mathbf{L} \mathbf{F}_{2,1,0} [\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_q^{\mathbf{s}}] + \\ & \frac{3}{4} \overline{\mathbf{y}}_u^{\text{pr}} \overline{\mathbf{y}}_u^{\text{st}} \mathbf{y}_u^{\text{pt}} \mathbf{y}_u^{\text{sr}} \mathbf{L} \mathbf{F}_{3,1,-1} [\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_q^{\mathbf{s}}] + \frac{1}{12} \mathbf{g}_1^2 \overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} \mathbf{L} \mathbf{F}_{2,2,0} [\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_u^{\mathbf{r}}] + \\ & \frac{3}{4} \mathbf{g}_2^2 \overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} \mathbf{L} \mathbf{F}_{3,1,0} [\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_u^{\mathbf{r}}] - \frac{1}{24} \overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} (\mathbf{g}_1^2 + 9 \mathbf{g}_2^2) \mathbf{L} \mathbf{F}_{3,2,-1} [\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_u^{\mathbf{r}}] - \\ & \frac{3}{4} \mathbf{g}_2^2 \overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} \mathbf{L} \mathbf{F}_{4,1,-1} [\mathbf{m}_q^{\mathbf{p}}, \mathbf{m}_u^{\mathbf{r}}] - \frac{1}{12} \mathbf{g}_1^2 \overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} \mathbf{L} \mathbf{F}_{3,1,0} [\mathbf{m}_u^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{p}}] - \\ & \frac{1}{12} \mathbf{g}_1^2 \overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} \mathbf{L} \mathbf{F}_{3,2,-1} [\mathbf{m}_u^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{p}}] - \frac{1}{4} \mathbf{g}_1^2 \overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} \mathbf{L} \mathbf{F}_{4,1,-1} [\mathbf{m}_u^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{p}}] + \\ & \frac{1}{2} \mathbf{g}_1^2 \overline{\mathbf{a}}_u^{\text{pr}} \mathbf{a}_u^{\text{pr}} \mathbf{L} \mathbf{F}_{5,1,-2} [\mathbf{m}_u^{\mathbf{r}}, \mathbf{m}_q^{\mathbf{p}}] - \frac{1}{16} \mathbf{g}_1^4 \mathbf{L$$